



University College Dublin

UCD | UAV 2025 | Team

X-Plane Handbook



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Introduction

This handbook outlines key characteristics of the UAV required to operate it in X-Plane, and instructions on how to operate the simulated version of the UAV replicating real world characteristics and control responses, as required by Annex D.6 in the IMechE UAS Challenge Competition Rules 2025. The UAV is a twin-motor, fixed-wing UAV developed for autonomous missions, however, it requires manual control in X-Plane. It features a 2.8m wingspan and is 10kg at takeoff. The UAV was developed in the latest version of X-Plane 11 (X-Plane 11.55 at the time of writing).

Aircraft Systems & Characteristics

Propulsion

The UAV uses two HobbyWing 4120 500KV brushless motors, which are simulated in X-Plane with equivalent electric motors of 3 horsepower.

Propeller

Each motor drives a 16x6" carbon fibre propeller. This was modelled in X-Plane by modifying propeller characteristics such as prop angle of attack, ellipse fraction, root and tip chord and rib chord until a desirable model was reached according to the manufacturer's specifications. The propellers have a maximum design rpm of 9000.

Wings and Control Surfaces

Main Wing - Semi-length = 1.4m - Root Chord = 0.34m - Tip Chord = 0.34m - Airfoil = Clark Y - Wing flex damping = 3.00	Vertical Stabiliser (Rudder) - Semi-length = 0.48m - Root Chord = 0.42m - Tip Chord = 0.25m - Airfoil = NACA 0009
Horizontal Stabiliser (Elevator) - Semi-length = 0.52m - Root Chord = 0.25m - Tip Chord = 0.25m - Airfoil = NACA 0014	Aileron - Semi-length = 0.323m - Root Chord = 0.1088m - Tip Chord = 0.1088m - Note: Also acts as speed brakes

Power Source

The UAV uses a 6S 9000mAh LiPo as the main power source and a 7.4V 2200mAh 2S LiPo as a secondary battery. As discussed in the limitations section of X-Plane, the secondary redundant battery had to be modelled as a duplicate of the main 6S LiPo battery. The motors and avionics are all connected to the main power source through bus #1, while the second battery acts as redundancy for essential components in case of a failure during flight.

Weight and Payload

The total UAV weight adds up to 10kg. The empty UAV weight is estimated to be 7.3kg and a payload of 2.7kg is expected to be used. The payload is modelled as a jettisonable water mass at the centre of gravity to minimise changes in the flight characteristics when the payload is released.

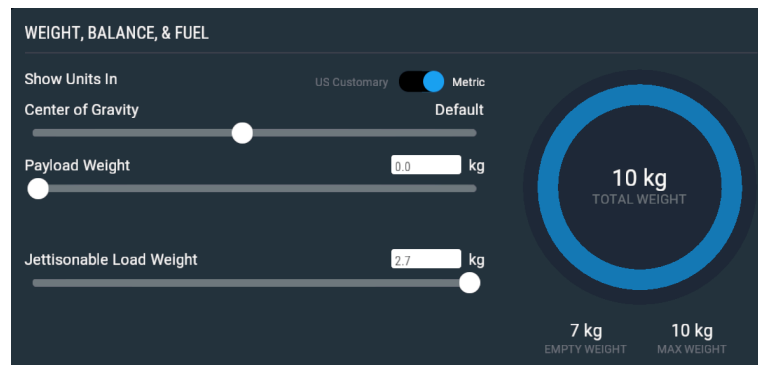
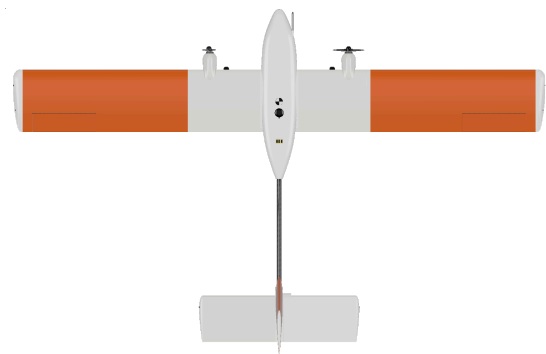


Image of the weight and balance configuration menu for the UAV. The empty weight of the UAV is 7.3kg, and the jettisonable water mass is 2.7kg.

Centre Of Gravity

The Centre of Gravity (CG) position is measured from the leading edge of the main wing. The static margin given by the chosen CG position is 15%.

Target CG position	0.173	m
Most forward CG position	0.156	m
Most aft CG position	0.19	m



Flight Control Surfaces

- Full-moving horizontal stabiliser (elevator).
- Full-moving vertical stabiliser (rudder).
- Ailerons on either end of the main wings. These also act as speed brakes when landing.

X-Plane Limitations

Batteries

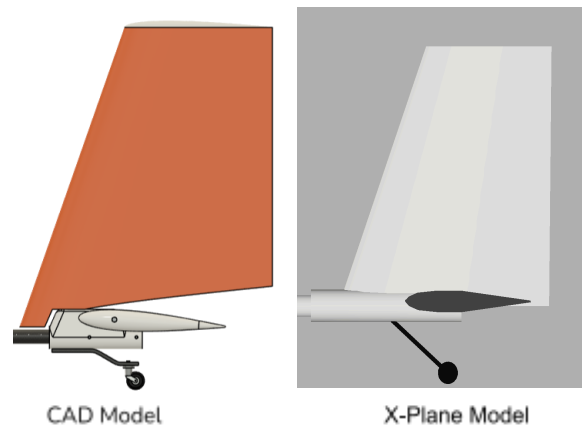
There were limitations to creating a second redundant battery with different characteristics from the main battery. Therefore, the secondary battery had to be identical to the main 6S 9000mAh LiPo battery. The main battery is defined as battery #1 and the secondary redundant battery is defined as battery #2. Battery #1 connects to bus #1 and battery #2 is connected to bus #2.

Landing Gear

The landing gears had to be extended slightly longer in X-Plane's Plane Maker due to limitations on the accuracy of these shorter-end landing gears. This was done in order to achieve a 7-degree angle of attack on take-off with all wheels touching the ground.

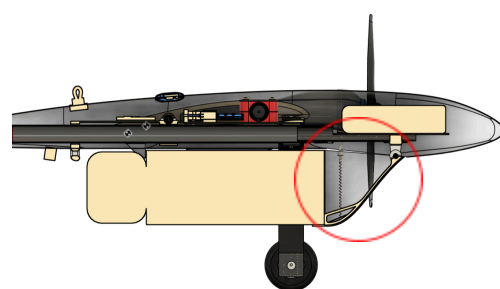
Vertical Stabiliser

There were limitations in the design of the vertical stabiliser also known as the rudder in Plane Maker. It was not possible in Plane Maker to create the distinctive undercut in the all-moving vertical stabiliser. The purpose of this undercut is to ensure that the rudder does not interfere with the horizontal stabilisers, however, this does not affect the flight in X-Plane.



Aero Flap

Our innovative design feature known as the aero flap was found to be difficult to model in X-Plane due to limitations in control logic and animation sequencing. The purpose of the aero flap is to provide an aerodynamic cover to the payload box while it is attached to the plane and retract into the fuselage once the payload is released. The aerodynamic advantages of the aero flap could not be simulated in X-Plane.



X-ray image of the UAV fuselage CAD, with the aero flap highlighted in the red circle.

Flight Procedure

This section provides details on how to operate and fly the UAV in X-Plane, including key speed markers and tips on aircraft control.

Takeoff

- Gradually increase the throttle to 100% while pulling back on the nose.
- Takeoff at roughly **28 KIAS** (14.5 m/s). Once the plane takes off, level the plane out and then maintain a 7 degree angle of attack (AOA) to climb upwards.
- Maintain control using ailerons and rudder.

Climb

- Maintain a steady climb at roughly **23 - 29 KIAS** (12-15 m/s)
- Climb at roughly 7 degrees AOA.

Cruise

- Maintain a cruising speed of **35 KIAS** (18 m/s) adjusting the throttle accordingly.

Landing

- Align with the landing area and begin the descent at **27 KIAS** (14 m/s)
- Use the throttle to control the rate of descent.
- Flare just before touchdown.
- Set throttle to 0% at a height of 5m, just before touchdown.
- Touchdown speed should be close to **23 KIAS** (12 m/s)
- When the UAV has landed, deploy ailerons as speed brakes. They should deflect to 30 degrees. Also manually deflect the elevator to a maximum angle of deflection.
- Maintain direction with the rudder while on the ground.

NOTE: This UAV does not include flaps. The approach and landing relies on throttle and pitch control.